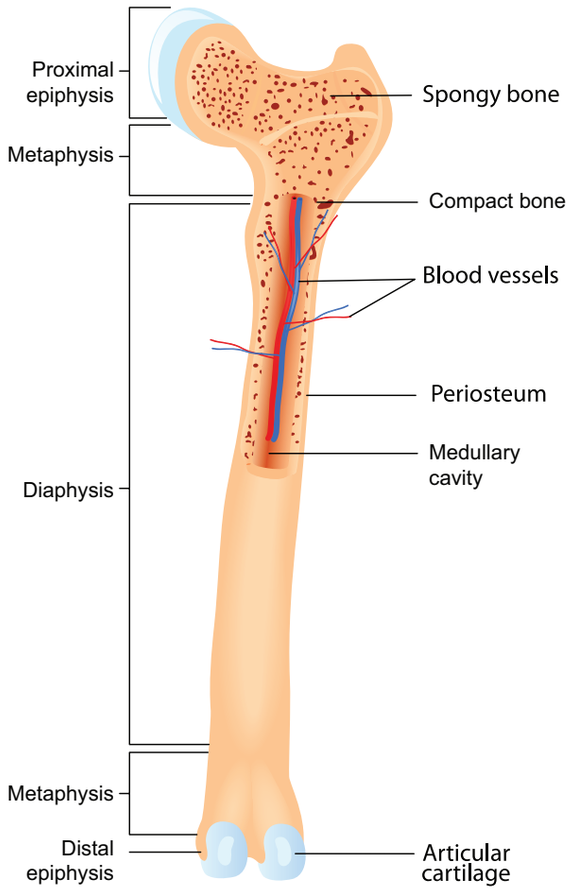


Figure 9.10 Structure of a long bone

9.10 Types of joints

Joints are essential for all types of movements performed by the bony parts of the body. The joints are points of contact (Figure 9.11) between bones.

Sometimes they are playing a protective role in the process. Force generated by the muscles are used to carry out the movement through joints which helps human functional activity of daily living and ambulation. The joint acts as a fulcrum of a lever.

(i) **Fibrous joints or Synarthroses:** They are immovable fixed joints in which no movement between the bones is possible. Sutures of the flat skull bones are fibrous joints.

(ii) **Cartilaginous joints or Amphiarthroses:** They are slightly movable joints in which the joint surfaces are separated by a cartilage and slight movement is only possible. E.g., Joints of adjacent vertebrae of the vertebral column.

(iii) Synovial joints or Diarthroses joints: They are freely movable joints, the articulating bones are seperated by a cavity which is filled with synovial fluid.

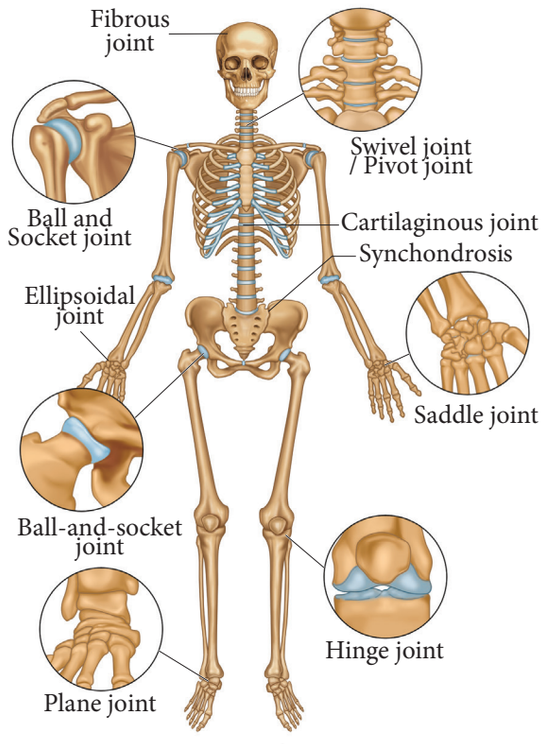


Figure 9. 11 Types of joints

Pivot joint	between atlas and axis
Plane/gliding joint	between the carpals
Saddle joint	between the carpal and metacarpal
Ball and socket joint	between humerus and pectoral girdle
Hinge joint	knee joint
Condyloid or Angular or Ellipsoid	between radius and carpal

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